COMPTIA SECURITY+ SY0-701 | DOMAIN 3 · SECURITY ARCHITECTURE (18%)

Security+ 3.4 · Backups

Mestre CyberSec · **Lesson 3.4 — Backups** (12:16) · Assistir Vídeo no YouTube ↗ · Treino Interativo: edusouzamarques.github.io/cybersec

Executive Lesson Summary: Frequência (por hora/dia/semana) + retention (quantos backups mantem); incremental (desde o último backup); differential (desde o último FULL). Snapshot: estado instantâneo. Replicação: tempo real. Regra 3-2-1: 3 cópias, 2 mídias, 1 offsite.

💡 Key Exam Highlights & Critical Concepts

- Incremental: desde o último backup (rápido, muitos arquivos small)
- Differential: desde o FULL (menos rápido, mais robusto)
- 3-2-1: 3 cópias, 2 mídias, 1 offsite; backup offline sobrevive ransomware

🎯 Exam Tips & Typical CompTIA Scenarios

- 'copies data since the last full backup'=differential
- '3 copies, 2 media types, 1 offsite'=3-2-1 rule

🚨 Beyond the Video: Official Exam Concepts, Protocols & Traps (Section 3.4 — Backups)

Why is this advanced coverage required for SY0-701?: Video lectures provide foundational concepts, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

• Full Backup

A complete copy of every file and object in the selected dataset at the time of backup. It is the baseline for differential/incremental backups, the fastest to restore, but the slowest to create and the most storage-intensive.

⚠ ***Exam Trap:*** *Questões citam o full como 'baseline' ou exige restauração com apenas 1 conjunto de dados. Distinção-chave: diferencial e incremental SEMPRE partem de um full — se não pedirem o próprio full, ele é o ponto de partida.*

• Incremental Backup

Backs up only data that changed since the LAST BACKUP of any type (full or incremental). Fastest to create with the smallest files, but a restore requires the full backup plus every incremental in the correct sequence.

⚠ ***Exam Trap:*** *Gatilho: 'data since the last backup' (de qualquer tipo) = INCREMENTAL. Não confunda: se a questão disser 'since the last FULL backup', é DIFFERENTIAL, não incremental.*

• Differential Backup

Backs up all data changed since the last FULL backup, accumulating data with each run. Restore needs only the full backup plus the latest differential, but each differential grows larger over time.

⚠ ***Exam Trap:*** *Gatilho clássico da prova: 'copies data since the last full backup' = DIFFERENTIAL. Distinção: restaurar com 2 conjuntos (full + último diferencial) vs. incremental (full + cadeia inteira de backups).*

• Snapshot

An instantaneous, point-in-time capture of the state of a system, VM, or storage volume, created in minutes with minimal impact. It is not a complete backup — it depends on the original volume and only tracks changes made after the capture.

⚠ ***Exam Trap:*** *'Instantaneous state / point in time' = SNAPSHOT. Distinção-chave: snapshot congela um instante (e ainda depende do volume original); replicação copia continuamente em tempo real; backup completo é independente.*

• Replication

Real-time (or near real-time) copying of files and data between devices or sites so that both locations hold synchronized copies. Provides high availability and redundancy, but corruption or ransomware is also replicated

instantly.

⚠️ ***Exam Trap:*** *'Real-time copy' = REPLICATION. Distinção-chave: replicação é cópia viva e contínua — NÃO protege sozinha contra ransomware (a cópia fica igualmente criptografada); snapshot/backup em pontos no tempo sim.*

• 3-2-1 Backup Rule

Best-practice backup strategy: keep at least 3 copies of the data, on 2 different media/storage types, with 1 copy held offsite (or offline/cloud). Protects against fire, flood, theft, and ransomware.

⚠️ ***Exam Trap:*** *Match direto no exame: '3 copies, 2 media types, 1 offsite' = REGRA 3-2-1. Cuidado com ordens invertidas nas distratoras (ex.: '2 offsite') — a sequência exata 3-2-1 é o que separa a resposta correta.*

• Backup Frequency and Retention

Backup policy parameters: frequency defines how often backups run (hourly/daily/weekly), while retention defines how many copies are kept and for how long. Together they balance recovery needs against storage costs.

⚠️ ***Exam Trap:*** *Separe os dois conceitos para não errar: FREQUÊNCIA = 'por hora/dia/semana' (quando rodar); RETENÇÃO = 'quantos backups manter' (por quanto tempo). Questões misturam os dois na mesma política de backup.*

• Offline Backup (Cold/Air-Gapped)

A backup copy kept disconnected from production systems and networks (tape, removable media, or air-gapped), making it unreachable by attackers and ransomware. An offline or cloud copy is also part of the 3-2-1 rule.

⚠️ ***Exam Trap:*** *Cenário típico: ransomware criptografa os backups acessíveis na rede → a cópia que sobrevive é a OFFLINE/air-gapped. Distinção crítica: OFFSITE (fora do local físico) ≠ OFFLINE (desconectada da rede) — uma cópia pode ser as duas coisas.*

🏛️ Conceptual Deep Dive, Real-World Analogies & GRC Analysis

📌 Backup Types: Full vs Incremental vs Differential

🏠 **Real-World Analogy:** Imagine a home ledger tracking family expenses. On Sunday night, you take a full-resolution photograph of EVERY ledger page (Full Backup)—time-consuming and consumes storage. Under the INCREMENTAL strategy: on Monday night, you jot down on a tiny sticky note only Monday's new expenses; on Tuesday night, on a second sticky note only Tuesday's new expenses. The daily backup is lightning fast! But if the ledger is lost on Wednesday, restoration requires the Sunday photo + Monday sticky note + Tuesday sticky note; lose Monday's note, and the recovery chain is broken! Under the DIFFERENTIAL strategy: on Monday night, you record everything added since Sunday; on Tuesday night, you record on a fresh page EVERYTHING added since Sunday (Monday + Tuesday accumulated). Daily backups take progressively longer, but restoring on Wednesday requires only two files: the Sunday photo + the latest Tuesday page.

🔍 **Root Cause & Background:** Executing daily full backups across enterprise petabyte arrays creates unsustainable network saturation and storage expense. Choosing Incremental vs Differential represents an explicit trade-off between nightly backup window speeds and critical emergency restoration duration (RTO).

🛡️ **GRC Auditor Perspective:** Auditing automated backup job logs; inspecting the state of the filesystem 'Archive Bit' (which Incremental clears/resets, while Differential leaves intact); and executing scheduled sample restore exercises to prevent discovering corrupted media during live disasters.

⚠️ **How CompTIA Tests This:** Exam trigger: 'fastest backup execution time, but slowest and most complex restore requiring Full plus all intervening daily increments' = Incremental Backup. Trigger: 'faster restore requiring only Full plus the latest single cumulative archive, with increasing daily backup windows' = Differential Backup.

📌 The Golden 3-2-1 Backup Rule

🏠 **Real-World Analogy:** Consider your family house deed or child's birth certificate. The 3-2-1 rule dictates: maintain 3 total copies (the physical original deed, a notarized physical duplicate, and a high-resolution scanned digital PDF); stored across 2 different media formats (cellulose paper in a file cabinet and digital blocks on an encrypted flash drive); and 1 copy located off-site, such as an encrypted cloud storage repository or a bank safe deposit box in another city. If your house suffers a catastrophic fire, your deed survives.

🔍 **Root Cause & Background:** Localized physical catastrophes (structural fires, burst sprinkler pipes, burglary) and automated malware simultaneously incinerate production hosts and the secondary USB hard drive left plugged into the same server rack.

🛡️ **GRC Auditor Perspective:** Auditing data protection repositories to verify 3 distinct data copies, confirming 2 dissimilar media types (flash disk arrays + LTO magnetic tape / immutable object storage), and validating sufficient geographical distance for the 1 off-site node.

⚠️ **How CompTIA Tests This:** Exam trigger: 'industry standard data protection framework mandating 3 total copies of critical business records, across 2 different storage media types, with at least 1 replica maintained off-site' = 3-2-1 Backup Rule.

📌 Snapshots vs Independent Backups

🏠 **Real-World Analogy:** A plastic ribbon bookmark placed inside a heavy textbook while studying (Snapshot). In one instant, you mark page 214, allowing you to flip back immediately if you lose your train of thought. But remember: if a fire destroys the textbook, the pages and your plastic bookmark burn together into ash! A true BACKUP is a duplicate textbook stored in a fireproof cabinet across town.

🔍 **Root Cause & Background:** Junior system administrators mistakenly treat virtual machine point-in-time snapshots as full backups. When the underlying SAN storage pool suffers filesystem corruption or hardware failure, the VM and all associated snapshot delta chains are destroyed.

🛡️ **GRC Auditor Perspective:** Enforcing explicit policies limiting snapshot retention lifecycles (mandating deletion within 24 to 72 hours following system patching); auditing automated daily backup exports to isolated secondary storage pools.

⚠️ **How CompTIA Tests This:** Exam trigger: 'instantaneous point-in-time capture of a virtual machine state suited for immediate rollback during software updates, dependent on host storage and insufficient as standalone backup' = Snapshot.

📌 Data Immutability & WORM Storage Against Ransomware

🏠 **Real-World Analogy:** Carving an ancient royal decree into solid granite with a hammer and chisel (WORM - Write Once, Read Many). Once chiseled into the stone, billions of citizens can inspect the text endlessly, but no one—not even the master sculptor, the king, or an invading warlord—can erase, overwrite, or sand down a single syllable for the next decade.

🔍 **Root Cause & Background:** Modern ransomware strains proactively target corporate backup repositories first. Attackers compromise enterprise domain administrator credentials to purge backup archives or encrypt catalog databases, guaranteeing the victim cannot recover without paying extortion ransoms.

🛡️ **GRC Auditor Perspective:** Mandating cloud storage Object Lock in strict Compliance Mode (preventing even root cloud credentials from deleting objects prior to retention expiration); auditing physical LTO tape cartridges configured with hardware-enforced WORM.

⚠️ **How CompTIA Tests This:** Exam trigger: 'safeguarding corporate backup stores against malicious ransomware encryption or administrative tampering by enforcing a write-once immutable state' = Immutability / WORM.

📌 Offline Backup & Air Gap (Physical Network Isolation)

🏠 **Real-World Analogy:** A cassette tape containing family recordings that you physically eject from the deck, seal in a padded aluminum case, and lock inside a subterranean vault with zero power cords, Ethernet cables, or Wi-Fi signals within miles. No cyber criminal on Earth can hack into that tape because there is no physical or electronic bridge connecting it to cyberspace.

🔍 **Root Cause & Background:** Online network-attached backup storage (hot repos) is readily compromised once an adversary gains Domain Administrator credentials, enabling automated scripts to systematically wipe out network shares and shadow copies.

🛡️ **GRC Auditor Perspective:** Auditing physical tape custody transfer logs with secure transport logistics vendors (e.g., Iron Mountain); verifying mechanical tape ejection procedures upon job completion ensuring complete physical Air-Gapping.

⚠️ **How CompTIA Tests This:** Exam trigger: 'maintaining backup media in a state physically isolated and completely disconnected from enterprise network fabrics to mitigate remote compromise' = Offline / Air-Gapped Backup.

⚡ QUICK-REFERENCE EXAM CHEAT SHEET

Typical Exam Scenario / Practical Situation	Technical Concept & Target Answer
Fastest nightly backup process capturing solely data modified since the most recent backup of any kind	**Incremental Backup**
Expedited restoration requiring exclusively the baseline full backup plus the latest cumulative archive	**Differential Backup**
Recommended framework maintaining 3 copies of business data, across 2 distinct media types, with 1 copy off-site	**3-2-1 Backup Rule**
Instantaneous point-in-time virtual machine capture utilized for immediate rollback prior to applying patches	**Snapshot**
Preventing backup files from deletion or ransomware encryption by enforcing write-once immutable locks	**Immutable Storage / WORM**
Magnetic tape physically detached from network circuits and locked in a vault to thwart remote hackers	**Offline / Air-Gapped Backup**

✅ Authentic Practice Questions (10 Real Exam Scenarios)

Scenario-based exam questions. Complete explanations and distractor analysis are provided in the answer key below.

QUESTION 1 · EXAM SCENARIO

A database administrator performs a full backup every Sunday. On Monday through Saturday, each backup job stores only the blocks that changed since the previous day's backup. Restoring the database to Saturday requires the Sunday full backup and every daily backup in sequence. Which backup method is being used?

A) Full backup

B) Incremental backup

C) Differential backup

D) Snapshot backup

QUESTION 2 · EXAM SCENARIO

An organization runs a full backup every Sunday. On Monday through Saturday, each backup job stores all files changed since Sunday, so the Saturday backup is larger than the Monday backup. To restore the latest data, administrators need only the Sunday full backup and the Saturday backup. Which backup method is in use?

A) Incremental backup

B) Snapshot backup

C) Differential backup

D) Replication

QUESTION 3 · EXAM SCENARIO

Before applying a critical patch to a production virtual machine, an administrator needs a point-in-time state that can be reverted quickly if the patch fails. The state must capture memory, disk, and configuration at that moment without performing a traditional file-level full backup. Which technology best meets this need?

A) Snapshot

B) Differential backup

C) Incremental backup

D) Offsite replication

QUESTION 4 · EXAM SCENARIO

A financial services company requires that transactions written to its primary data center appear at a secondary site within seconds. The recovery strategy must minimize data loss if the primary site fails, and it must continuously copy changes rather than wait for a nightly backup window. Which strategy best matches this requirement?

A) Weekly full backup

B) Daily differential backup

C) Monthly archive backup

D) Replication

QUESTION 5 · EXAM SCENARIO

An audit finds that an organization keeps three backup copies, but all are stored on the same NAS appliance in the same server room. Which change best aligns the backup architecture with the 3-2-1 rule?

A) Increase backup frequency to every hour

B) Enable deduplication on the NAS

C) Store one copy offsite on different media

D) Convert full backups to incremental

QUESTION 6 · EXAM SCENARIO

A company's backups are stored on a network share that is always mounted and accessible with domain administrator credentials. A ransomware incident encrypts both production data and the mounted backup share. Which control best protects future backups from a similar ransomware impact?

A) Increase retention from 30 to 90 days

B) Maintain an offline or air-gapped immutable backup copy

C) Switch from differential to incremental backups

D) Configure RAID 6 on the backup storage

QUESTION 7 · EXAM SCENARIO

A legal team requires the ability to restore financial records from any day within the past seven years. Storage capacity is limited, so administrators must define how long each backup set is preserved before deletion. Which backup policy setting directly addresses this requirement?

A) Retention period

B) Backup frequency

C) Compression ratio

D) Replication interval

QUESTION 8 · EXAM SCENARIO

An e-commerce platform has a recovery point objective of 15 minutes. Its current backup schedule runs incremental backups every four hours, and the business cannot accept losing more than 15 minutes of orders. Which adjustment most directly addresses the RPO gap?

A) Extend backup retention to 180 days

B) Replace incremental backups with differential backups

C) Enable encryption for backup tapes

D) Reduce the backup interval to 15 minutes or less

QUESTION 9 · EXAM SCENARIO

A server fails on Thursday evening. The backup schedule is a full backup on Sunday and an incremental backup every Monday through Thursday. Administrators need to restore the most recent data possible. Which restore set is required?

A) Sunday full backup only

B) Thursday incremental backup only

C) Sunday full backup plus Monday, Tuesday, Wednesday, and Thursday incrementals

D) Sunday full backup plus Thursday incremental only

QUESTION 10 · EXAM SCENARIO

A file server is protected by a full backup every Sunday and a differential backup every Monday through Saturday. After a disk failure on Thursday, administrators must restore the latest available data. Which restore set should they use?

A) Sunday full backup plus all differential backups from Monday through Thursday

B) Sunday full backup plus Thursday differential backup

C) Thursday differential backup only

D) Monday through Thursday differential backups only

ANSWER KEY & COMPREHENSIVE DISTRACTOR EXPLANATIONS

Question 1: Option B — Incremental backup

Incremental backup captures only changes made since the most recent backup of any type, so a restore requires the last full backup plus every incremental backup in order. Full backup copies all data every time. Differential backup captures changes since the last full backup. Snapshot captures a point-in-time state but is not the file-level chain described here.

Question 2: Option C — Differential backup

Differential backup stores all changes since the last full backup, so each differential grows until the next full backup. Restore requires the last full backup plus the latest differential. Incremental would require every incremental since the full. Snapshot is a point-in-time state. Replication continuously copies data rather than producing a daily backup set.

Question 3: Option A — Snapshot

A snapshot records the state of a system or virtual machine at a specific point in time and allows fast rollback before or after changes such as patching. Differential and incremental backups are file-level backup methods that may take longer to restore. Offsite replication copies data to another location but is not primarily used to revert a patch state on the same VM.

Question 4: Option D — Replication

Replication continuously copies changes to another system or site in near real time, helping meet a very low recovery point objective. Weekly full, daily differential, and monthly archive backups operate on scheduled intervals and would lose more data if the primary site failed. They do not provide the required near-real-time copy.

Question 5: Option C — Store one copy offsite on different media

The 3-2-1 rule requires at least three copies of data, on two different media types, with one copy offsite. Keeping all backups on one NAS in one room does not satisfy media diversity or offsite protection. Increasing frequency, enabling deduplication, or changing backup type does not address the lack of offsite and separate media.

Question 6: Option B — Maintain an offline or air-gapped immutable backup copy

An offline, air-gapped, or immutable backup copy cannot be reached or altered by ransomware that is spreading through the network. Retention changes do not prevent encryption of online backups. Incremental backups are still mounted and accessible, so they can be encrypted. RAID provides availability against disk failure, not protection against ransomware.

Question 7: Option A — Retention period

Retention period defines how long backup sets are kept before deletion and directly supports the seven-year restore requirement. Backup frequency determines how often backups are created and affects RPO. Compression ratio affects storage use. Replication interval affects how often copies are transferred to another location.

Question 8: Option D — Reduce the backup interval to 15 minutes or less

The recovery point objective defines the maximum acceptable data loss, so backup frequency must be at least as often as the RPO. Running backups every four hours allows up to four hours of data loss, so reducing the interval to 15 minutes or less directly closes the gap. Retention, switching to differential, or encryption does not change how much recent data could be lost.

Question 9: Option C — Sunday full backup plus Monday, Tuesday, Wednesday, and Thursday incrementals

Incremental backups contain only changes since the previous backup, so a full restore requires the last full backup plus every incremental backup in sequence. Using only the Sunday full backup loses later changes. Using only Thursday incremental is incomplete. Sunday full plus only Thursday incremental would be valid for a differential backup, not an incremental chain.

Question 10: Option B — Sunday full backup plus Thursday differential backup

Differential backups accumulate all changes since the last full backup, so the Thursday differential already includes changes from Monday through Thursday. Restoring the Sunday full backup and the latest differential restores the latest data. Restoring all differentials is unnecessary and may be slower. Restoring only differentials without the full backup is incomplete.